

Geodesic equations and connection coefficients

9

I know now that if I break my neck by falling off a cliff, my death is not to be blamed on the force of gravity (what does not exist is necessarily guiltless), but on the fact that I did not maintain the first curvature of my world-line, exchanging its security for a dangerous geodesic.

John Lighton Synge (1897–1995)

Synge’s words, quoted above, remind us that our new geometric perspective on gravity motivates us to think about the curvature of our world line.¹ In the last two chapters, we saw that how basis vectors change as we move through spacetime is reflected in the connection coefficients $\Gamma^\mu_{\alpha\beta}$ which feature in the geodesic equation, which is the equation of motion for a particle in free fall. The connection coefficients are important, not only for the role they play in the geodesic equation, but because they tell us about the **curvature** of spacetime itself. In this chapter, we describe a method to extract the connection coefficients. As shown in Fig. 9.1, the idea is to input the metric and outputs the connection. The most important point of this chapter is the following.

The metric field of spacetime, via the line element $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$, generates the geodesics that freely falling particles follow and, therefore, the connection coefficients.

9.1 Finding connection coefficients

Let’s formulate our method. The interval between spacetime points a and b can be written as the integral²

$$s = \int_a^b d\lambda \left| g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right|^{\frac{1}{2}}. \quad (9.3)$$

We use the Euler–Lagrange equations on the integrand to find the equations of motion. We saw in the last chapter that our expressions can be simplified by choosing length parametrization after the first set of derivatives have been taken and this is necessary to interpret s as the interval between spacetime points.³ The equations of motion can be written in the form of the geodesic equation

$$\frac{d^2 x^\mu}{d\lambda^2} + \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} \Gamma^\mu_{\alpha\beta} = 0. \quad (9.4)$$

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¹A world line which, we very much hope, will avoid any cliff falls.

$$g \implies ds^2 \implies \Gamma$$

Fig. 9.1 The metric generates the connection coefficients.

²For spacelike curves we have $ds^2 > 0$, and we can write the interval as

$$\Delta l = \int_a^b d\lambda \left(g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right)^{\frac{1}{2}}. \quad (9.1)$$

This is the proper length along the curve. Massive particles traverse timelike curves which have $ds^2 < 0$. We can write the timelike interval

$$\Delta\tau = \int_a^b d\lambda \left(-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right)^{\frac{1}{2}}. \quad (9.2)$$

This equation gives us the proper time that elapses for an observer travelling along the world line.

³We described the need for this in the previous chapter. For massive particles, which follow timelike geodesics, length parametrization, involving setting $\lambda = \tau$, is also needed to ensure that the velocity vector $\mathbf{u}$, which is the tangent of the particle’s world line, is constrained according to $\mathbf{u} \cdot \mathbf{u} = -1$.

↪ The rest of this chapter goes through in detail how to extract connection coefficients and calculate geodesics. A reader impatient to get on to the heart of general relativity can skip the rest of this chapter on first reading.

By comparing the equations of motion we can simply read off the connection coefficients.

The method formalizes the method used in the examples in the previous chapter. It can be summarized as follows:

Step I: From the metric line element ds^2 , write a parametrized expression for the spacetime interval $s = \int L d\lambda$, where λ is the parameter.
Step II: Calculate $\frac{\partial L}{\partial(\frac{dx^\mu}{d\lambda})}$ and $\frac{\partial L}{\partial x^\mu}$.
Step III: Choose length parametrization such that $L = 1$.
Step IV: Calculate $\frac{d}{d\lambda} \frac{\partial L}{\partial(\frac{dx^\mu}{d\lambda})}$ and insert the values into the E-L equations.
Step V: Read off the connection coefficients, remembering that $\Gamma^\mu_{\alpha\beta} = \Gamma^\mu_{\beta\alpha}$.

We shall work through a number of examples demonstrating how to extract connection coefficients. In each case, we start with a metric and end with connection coefficients.⁴

⁴These geometries reappear in several of the exercises and examples later in the book.

Tips for these calculations:

■ To save on writing, in step I it's sometimes useful to write $\frac{dx^\mu}{d\lambda}$ as $\dot{x}^\mu$.

■ In step IV, we often employ the chain rule, noting that $\frac{d}{d\lambda} f(x^\mu) = \frac{\partial f(x^\mu)}{\partial x^\mu} \frac{dx^\mu}{d\lambda}$.

■ In step V, equations of motion of the form $\ddot{x}^\mu + 2F\dot{x}^\alpha x^\beta = 0$ for $\alpha \neq \beta$ yield $\Gamma^\mu_{\alpha\beta} = F$, owing to the summation convention in the geodesic equation.

■ The length parametrization condition $L = 1$ is often a useful, additional constraint when solving the equations of motion.

Example 9.1

Let's try the two-dimensional space on the surface of a unit sphere. The interval (step I) is

$$\begin{aligned} s &= \int (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)^{\frac{1}{2}} \\ &= \int d\lambda \left[\left(\frac{d\theta}{d\lambda} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\lambda} \right)^2 \right]^{\frac{1}{2}}. \end{aligned} \quad (9.5)$$

Therefore, the metric has components $g_{\theta\theta} = 1$ and $g_{\phi\phi} = \sin^2 \theta$. We saw [in Exercise 8.1 from the last chapter] that (following steps II–IV) the equations of motion are

$$\begin{aligned} \frac{d^2\theta}{d\lambda^2} - \sin \theta \cos \theta \left(\frac{d\phi}{d\lambda} \right)^2 &= 0, \\ \frac{d^2\phi}{d\lambda^2} + 2 \cot \theta \frac{d\phi}{d\lambda} \frac{d\theta}{d\lambda} &= 0, \end{aligned} \quad (9.6)$$

and we read off the connection coefficients (step V)

$$\Gamma^\theta_{\phi\phi} = -\sin \theta \cos \theta, \quad \Gamma^\phi_{\theta\phi} = \cot \theta. \quad (9.7)$$

We can examine different curved surfaces, such as the parabolic space of the next example.

Example 9.2

A parabolic surface with line element $ds^2 = (1 + a^2 r^2) dr^2 + r^2 d\theta^2$ has interval (step I)

$$s = \int d\lambda \left[(1 + a^2 r^2) \left(\frac{dr}{d\lambda} \right)^2 + r^2 \left(\frac{d\theta}{d\lambda} \right)^2 \right]^{\frac{1}{2}}. \quad (9.8)$$

Step II of the method gives us

$$\begin{aligned}\frac{\partial L}{\partial(\frac{dr}{d\lambda})} &= \frac{1}{L}(1+a^2r^2)\frac{dr}{d\lambda}, & \frac{\partial L}{\partial(\frac{d\theta}{d\lambda})} &= \frac{1}{L}r^2\frac{d\theta}{d\lambda}, \\ \frac{\partial L}{\partial r} &= \frac{1}{L}\left[a^2r\frac{dr}{d\lambda} + r\left(\frac{d\theta}{d\lambda}\right)^2\right], & \frac{\partial L}{\partial\theta} &= 0.\end{aligned}\quad (9.9)$$

Now use length parametrization (step III) and find the equations of motion (step IV). Here's the first

$$\frac{d^2r}{d\lambda^2} + \frac{a^2r}{1+a^2r^2}\left(\frac{dr}{d\lambda}\right)^2 - \frac{r}{1+a^2r^2}\left(\frac{d\theta}{d\lambda}\right)^2 = 0. \quad (9.10)$$

And the second

$$\frac{d^2\theta}{d\lambda^2} + \frac{2}{r}\frac{dr}{d\lambda}\frac{d\theta}{d\lambda} = 0. \quad (9.11)$$

We read off the connection coefficients (step V)

$$\Gamma_{r\theta}^\theta = \frac{1}{r}, \quad \Gamma_{rr}^r = \frac{a^2r}{1+a^2r^2}, \quad \Gamma_{\theta\theta}^r = -\frac{r}{1+a^2r^2}. \quad (9.12)$$

As expected, these reduce down to the flat-plane connection coefficients in the case that $a = 0$.

We can examine more exotic spaces still, such as the interesting Poincaré half plane.⁵

⁵Henri Poincaré (1854–1912). The Poincaré half plane provides a model of hyperbolic geometry. See Exercise 9.8 for an introduction to the Poincaré half plane and Chapters 16 and 19 for more discussion of hyperbolic spaces.

Example 9.3

The Poincaré half plane has a metric

$$ds^2 = \frac{1}{r^2}(dr^2 + dx^2), \quad (9.13)$$

which is defined for $r > 0$. The interval (step I) is

$$s = \int d\lambda \left[\frac{1}{r^2} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{r^2} \left(\frac{dx}{d\lambda} \right)^2 \right]^{\frac{1}{2}}. \quad (9.14)$$

Put this through the standard machine (step II)

$$\begin{aligned}\frac{\partial L}{\partial(\frac{dr}{d\lambda})} &= \frac{1}{L}\frac{1}{r^2}\frac{dr}{d\lambda}, & \frac{\partial L}{\partial(\frac{dx}{d\lambda})} &= \frac{1}{L}\frac{1}{r^2}\frac{dx}{d\lambda}, \\ \frac{\partial L}{\partial r} &= -\frac{1}{L}\frac{1}{r^3}\left[\left(\frac{dr}{d\lambda}\right)^2 + \left(\frac{dx}{d\lambda}\right)^2\right], & \frac{\partial L}{\partial x} &= 0.\end{aligned}$$

Now use length parametrization (step III) so that we make $L = 1$. We then find

$$\begin{aligned}\frac{d}{d\lambda} \left(\frac{1}{r^2} \frac{dr}{d\lambda} \right) &= \frac{1}{r^2} \frac{d^2r}{d\lambda^2} - \frac{2}{r^3} \frac{dr}{d\lambda} \frac{dr}{d\lambda}, \\ \frac{d}{d\lambda} \left(\frac{1}{r^2} \frac{dx}{d\lambda} \right) &= \frac{1}{r^2} \frac{d^2x}{d\lambda^2} - \frac{2}{r^3} \frac{dr}{d\lambda} \frac{dx}{d\lambda}.\end{aligned}\quad (9.15)$$

We obtain equations of motion (step IV)

$$\begin{aligned}\frac{d^2r}{d\lambda^2} - \frac{2}{r} \frac{dr}{d\lambda} \frac{dr}{d\lambda} &= -\frac{1}{r} \left[\left(\frac{dr}{d\lambda} \right)^2 + \left(\frac{dx}{d\lambda} \right)^2 \right], \\ \frac{d^2x}{d\lambda^2} - \frac{2}{r} \frac{dr}{d\lambda} \frac{dx}{d\lambda} &= 0.\end{aligned}\quad (9.16)$$

From the first of these we obtain

$$\frac{d^2r}{d\lambda^2} - \frac{1}{r} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{r} \left(\frac{dx}{d\lambda} \right)^2 = 0. \quad (9.17)$$

We read off connection coefficients (step V)

$$\Gamma_{xr}^x = \Gamma_{rx}^x = -\frac{1}{r}, \quad \Gamma_{rr}^r = -\frac{1}{r}, \quad \Gamma_{xx}^r = \frac{1}{r}. \quad (9.18)$$

The geodesics themselves turn out to be circular arcs and are examined in Exercise 9.8.

9.2 The geodesic equation from the action

So far we have looked at a selection of special cases, evaluating spacelike intervals for space-only metrics with a (+++) signature.⁶ However, using the Euler–Lagrange equations, we should be able to derive the equation of motion for a massive particle in a general spacetime, once and for all, from the action

$$S = -m \int d\tau \left(-g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right)^{\frac{1}{2}}. \quad (9.19)$$

This action is proportional to the proper time interval along a world line parametrized by the proper time τ . We can therefore extremize the action using the same procedure as before. We already know what the answer must be: the geodesic equation from the previous chapter. However, this procedure will also provide a useful and simple formula for extracting the connection coefficients directly from the metric.⁷

⁶Lots more examples can be found in the exercises.

⁷Although the resulting expression is useful, it is often quicker in practice to use the five-point method to extract the coefficients directly from the action.

Example 9.4

We identify (step I)

$$L = -m \left(-g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right)^{\frac{1}{2}}. \quad (9.20)$$

Step II yields

$$\frac{\partial L}{\partial \dot{x}^\mu} = m g_{\mu\nu} \frac{dx^\nu}{d\tau} \frac{1}{L}, \quad (9.21)$$

along with a force term

$$\frac{\partial L}{\partial x^\nu} = \frac{m}{2} \frac{\partial g_{\mu\sigma}}{\partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\sigma}{d\tau} \frac{1}{L}. \quad (9.22)$$

Choose the parametrization such that $L = 1$ (Step III) and we find, at step IV, that

$$\frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}^\mu} = m \frac{d}{d\tau} \left(g_{\mu\nu} \frac{dx^\nu}{d\tau} \right). \quad (9.23)$$

The Euler–Lagrange equation then reads

$$\frac{d}{d\tau} \left(g_{\mu\nu} \frac{dx^\nu}{d\tau} \right) = \frac{1}{2} \frac{\partial g_{\mu\sigma}}{\partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\sigma}{d\tau}. \quad (9.24)$$

We can evaluate the left-hand side to find

$$\frac{d}{d\tau} \left(g_{\mu\nu} \frac{dx^\nu}{d\tau} \right) = g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{\partial g_{\mu\nu}}{\partial x^\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau}. \quad (9.25)$$

The Euler–Lagrange equation becomes

$$g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \left(\frac{\partial g_{\mu\nu}}{\partial x^\sigma} - \frac{1}{2} \frac{\partial g_{\mu\sigma}}{\partial x^\nu} \right) \frac{dx^\mu}{d\tau} \frac{dx^\sigma}{d\tau} = 0. \quad (9.26)$$

We now tidy up by noting that, since the metric components are symmetric,

$$\dot{x}^\mu \dot{x}^\sigma g_{\mu\nu,\sigma} = \frac{\dot{x}^\mu \dot{x}^\sigma}{2} (g_{\mu\nu,\sigma} + g_{\sigma\nu,\mu}), \quad (9.27)$$

which allows us to conclude (step V):

$$g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{1}{2} \left(\frac{\partial g_{\mu\nu}}{\partial x^\sigma} + \frac{\partial g_{\sigma\nu}}{\partial x^\nu} - \frac{\partial g_{\mu\sigma}}{\partial x^\nu} \right) \frac{dx^\mu}{d\tau} \frac{dx^\sigma}{d\tau} = 0. \quad (9.28)$$

We define the all-down-index connection coefficients $\Gamma_{\lambda\mu\sigma} = g_{\rho\lambda}\Gamma^\rho_{\mu\sigma}$, and then we have the following.⁸

$$\Gamma_{\lambda\mu\sigma} = \frac{1}{2} \left(\frac{\partial g_{\lambda\mu}}{\partial x^\sigma} + \frac{\partial g_{\lambda\sigma}}{\partial x^\mu} - \frac{\partial g_{\mu\sigma}}{\partial x^\lambda} \right). \quad (9.29)$$

The conclusion of this lengthy exercise⁹ is that, given only the metric, we can work out connection coefficients and have access to the equation of motion of the freely falling particle. Note, however, that since the geodesic equation applies beyond the timelike geodesics followed by massive particles, eqn 9.29 is a general, geometrical expression linking the metric with the connection coefficients.

Example 9.5

Consider the metric line element

$$ds^2 = \frac{1}{t^2} (dx^2 - dt^2). \quad (9.33)$$

This corresponds to a metric with components $g_{tt} = -1/t^2$ and $g_{xx} = 1/t^2$. The connection coefficients can be calculated using eqn 9.28, to yield

$$\Gamma_{ttt} = \frac{1}{t^3}, \quad \Gamma_{xxt} = \Gamma_{xtx} = -\Gamma_{txx} = -\frac{1}{t^3}. \quad (9.34)$$

Using $g^{tt} = -t^2$ and $g^{xx} = t^2$, we have $\Gamma^t_{tt} = \Gamma^t_{xx} = \Gamma^x_{xt} = -1/t$.

⁸In comma notation:

$$\Gamma_{\lambda\mu\sigma} = \frac{1}{2} (g_{\lambda\mu,\sigma} + g_{\lambda\sigma,\mu} - g_{\mu\sigma,\lambda}). \quad (9.29a)$$

⁹There are some other useful expressions that can be employed in calculations. One useful identity is

$$g_{\alpha\beta,\gamma} = \Gamma_{\alpha\beta\gamma} + \Gamma_{\beta\alpha\gamma}. \quad (9.30)$$

Another is

$$\Gamma^\alpha_{\beta\alpha} = \frac{\partial}{\partial x^\beta} \ln \sqrt{-g}, \quad (9.31)$$

where g is the determinant of the metric tensor. For a diagonal metric the following hold

$$\begin{aligned} \Gamma^\mu_{\nu\lambda} &= 0, \\ \Gamma^\mu_{\lambda\lambda} &= -\frac{1}{2g_{\mu\mu}} \frac{\partial g_{\lambda\lambda}}{\partial x^\mu}, \\ \Gamma^\mu_{\mu\lambda} &= \frac{\partial}{\partial x^\lambda} \left(\ln |g_{\mu\mu}|^{\frac{1}{2}} \right), \\ \Gamma^\mu_{\mu\mu} &= \frac{\partial}{\partial x^\mu} \left(\ln |g_{\mu\mu}|^{\frac{1}{2}} \right). \end{aligned} \quad (9.32)$$

Here $\mu \neq \nu \neq \lambda$ and we don't sum over repeated indices.

Chapter summary

- The connection coefficients may be extracted using a simple routine based on extremizing the action.
- The metric leads directly to the connection coefficients.

Exercises

(9.1) Using the methods described in the chapter, extract the connection coefficients for two-dimensional plane polar coordinates.

(9.2) The torus has a line element

$$ds^2 = (c + a \cos v)^2 du^2 + a^2 dv^2. \quad (9.35)$$

Show that we obtain the connection coefficients

$$\Gamma^v_{uu} = \frac{\sin v}{u} (c + a \cos v), \quad \Gamma^u_{uv} = -\frac{a \sin v}{(c + a \cos v)}. \quad (9.36)$$

(9.3) Consider the non-diagonal metric

$$ds^2 = du^2 + dv^2 + 2dudv \cos \theta(u, v). \quad (9.37)$$

Show that the non-zero connection coefficients are given by

$$\Gamma^u_{uu} = \frac{\cos \theta}{\sin \theta} \theta_{,u}, \quad \Gamma^v_{uu} = -\frac{1}{\sin \theta} \theta_{,u}, \quad \Gamma^u_{vv} = \frac{\cos \theta}{\sin \theta} \theta_{,v}, \quad \Gamma^v_{vv} = -\frac{1}{\sin \theta} \theta_{,v} \quad (9.38)$$

- (9.4) Consider Rindler spacetime with metric line element

$$ds^2 = -x^2 dt^2 + dx^2. \quad (9.39)$$

- (a) Show that the equations of motion are

$$2\dot{x}\ddot{t} + x\ddot{t} = 0, \quad \ddot{x} + x\dot{t}^2 = 0, \quad (9.40)$$

and extract the connection coefficients.

- (b) We can use the physical interpretation of the length parametrization to allow some insight into this spacetime. Define the velocity in the x -direction as $v = dx/dt$ and show that

$$\ddot{x} = -\frac{x}{x^2 - v^2}. \quad (9.41)$$

Now suppose that the particle does not have a velocity in the x -direction, so that $v = 0$. The equation of motion says that in order to have $v = 0$ in this spacetime we must have an observer undergoing a uniform acceleration $\ddot{x} = -1/x$, which diverges as $x \rightarrow 0$. We shall see this property again when we examine the spherically symmetric Schwarzschild geometry.

- (9.5) Consider the rotating-frame line element

$$ds^2 = -[1 - \Omega^2(x^2 + y^2)]dt^2 + dx^2 + dy^2 + dz^2 - 2\Omega y dx dt + 2\Omega x dy dt. \quad (9.42)$$

- (a) Find the matrix $g^{\mu\nu}$.

- (b) Compute the connection coefficients for the space described by this line element.

- (9.6) (a) Express the line element from the previous question in cylindrical polars.

- (b) Compute the connection coefficients in cylindrical polars.

- (9.7) The Schwarzschild metric gives an interval

$$s = \int d\lambda \left[e^{2\Phi} \left(\frac{dt}{d\lambda} \right)^2 - e^{2\Lambda} \left(\frac{dr}{d\lambda} \right)^2 - r^2 \left(\frac{d\theta}{d\lambda} \right)^2 - r^2 \sin^2 \theta \left(\frac{d\phi}{d\lambda} \right)^2 \right]^{\frac{1}{2}}, \quad (9.43)$$

where Φ and Λ are functions of r . Find the connection coefficients.

- (9.8) We can start to understand the space represented by the Poincaré half plane in Example 9.3 by computing its geodesics. The geometry is defined by its metric in the upper half plane, $r > 0$, only.

- (a) Consider the equation of motion $\ddot{x} - 2\dot{r}\dot{x}/r = 0$, where the dot indicates a derivative with respect to

the affine parameter λ . Show that this equation is solved by

$$\dot{x} = \frac{r^2}{a}, \quad (9.44)$$

where a is a constant.

- (b) Consider the length parametrization condition $L = 1$ and show that this yields

$$d\lambda = \frac{dt}{\sin t}, \quad (9.45)$$

where $r = a \sin t$.

- (c) Use these results to show that x is given by

$$x = -a \cos t + x_0, \quad (9.46)$$

where x_0 is a constant offset.

- (d) Argue that this shows that the geodesics are circular arcs, centred on $(x, r) = (x_0, 0)$ with radius a , as shown in Fig. 9.2.

- (e) Compute the length $\int d\lambda$ of a geodesic starting at $t = a$ and finishing at $t = b$. Use this to show that the length of a geodesic starting at $t = 0$ and ending at $t = \pi$ is infinite.

You can see why this is the case by considering a ruler of interval length Δs parallel to the x axis. Since r is constant we have $\Delta s = \Delta x/r$. As a result, rulers of an equivalent interval length Δs must have larger coordinate length Δx if they are at a larger height r , as shown in Fig. 9.2.

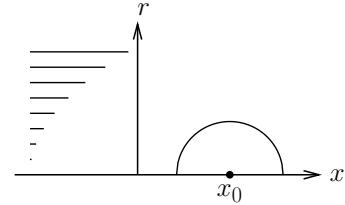


Fig. 9.2 The Poincaré half plane from Exercise 9.8 and Example 9.3. An example geodesic is shown on the right. On the left several lines of equivalent interval length Δs are shown.

- (9.9) We end up with the same geodesics if we extremize $L = \sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}$, and if we extremize $L = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$. Show this by finding the Euler-Lagrange equation for a function

$$L = F\left(\sqrt{g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}\right), \quad (9.47)$$

where $\dot{x}^\mu = \frac{dx^\mu}{d\lambda}$, λ is the proper length and F is any monotonic function.

This means we can equally well use a Lagrangian $L = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ which resembles the kinetic energy of a non-relativistic particle.

(9.10) Consider the moving-coordinate metric

$$ds^2 = -(1 - v^2) dt^2 + dx^2 + dy^2 + dz^2 - 2v dx dt. \quad (9.48)$$

By extremizing the world line, show that the geodesics are straight lines.

(9.11) Consider a two-dimensional space with metric line element

$$ds^2 = (1 + r)dr^2 + r^2 d\phi^2. \quad (9.49)$$

By computing the acceleration, determine whether the curve $r(\lambda) = (3\lambda/2)^{2/3} - 1$, $\phi(\lambda) = 0$, with λ an affine parameter, is a geodesic.